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Received: 16 October 2023 | Revised: 24 December 2023 | Accepted: 9 January 2024

Energy Conversion and Economics

DOI: 10.1049/enc2.12122

The Institution of Engineering and Technology WILEY

ORIGINAL RESEARCH

Carbon market risk estimation using quantum conditional generative adversarial network and amplitude estimation

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Funding information
National Natural Science Foundation of China, Grant/Award Numbers: 72331009, 72171206; Guangdong Power Grid Company, Grant/Award Number: 036000KC23090003(GDKJXM20231024); Shenzhen Key Lab of Crowd Intelligence Empowered Low-Carbon Energy Network, Grant/Award Number: ZDSYS20220606100601002; Shenzhen Natural Science Fund, Grant/Award Number: GXWD20231128112434001; Shenzhen Institute of Artificial Intelligence and Robotics for Society

Abstract
Accurately and efficiently estimating the carbon market risk is paramount for ensuring financial stability, promoting environmental sustainability, and facilitating informed decision-making. Although classical risk estimation methods are extensively utilized, the implicit pre-assumptions regarding distribution are predominantly contained and challenging to balance accuracy and computational efficiency. A quantum computing-based carbon market risk estimation framework is proposed to address this problem with the quantum conditional generative adversarial network-quantum amplitude estimation (QCGAN-QAE) algorithm. Specifically, quantum conditional generative adversarial network (QCGAN) is employed to simulate the future distribution of the generated return rate, whereas quantum amplitude estimation (QAE) is employed to measure the distribution. Moreover, the quantum circuit of the QCGAN improved by reordering the data interaction layer and data simulation layer is coupled with the introduction of the quantum fully connected layer. The binary search method is incorporated into the QAE to bolster the computational efficiency. The simulation results based on the European Union Emissions Trading System reveals that the proposed framework markedly enhances the efficiency and precision of value-at-risk and conditional value-at-risk compared to original methods.

KEYWORDS
carbon market, quantum amplitude estimation, quantum conditional generative adversarial networks, risk estimation

1 | INTRODUCTION

The international society has committed to a consensus on carbon emissions reduction in response to the severe challenges of rising global temperatures and environmental degradation [1]. The establishment of carbon markets has been recognised as an essential and flexible strategy to realise this collective vision. Among the carbon markets of various countries and

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regions, the European Union Emissions Trading System (EU ETS) is pivotal in mitigating greenhouse gas emissions through market mechanisms [2]. However, the EU carbon market is frequently challenged by price volatility owing to factors such as supply-demand dynamics, policy adaptations, and global economic condition changes, consequently amplifying the financial risks inherent to the carbon market. Imprecise risk estimates can result in significant financial losses for market participants. Thus, it is imperative to employ refined techniques to estimate carbon market risks accurately.

Existing research has explored three main methods to enhance the accuracy of market risk estimation: the time-series model, the capital asset pricing model, and Monte Carlo simulation. Each method corresponds to a different risk measure. Time-series models, when applied to historical data, can compute the volatility and variance of financial assets, offering investors an initial measure of market price or return fluctuations [3, 4]. The capital asset pricing model determines the beta coefficient [5, 6]. This coefficient measures the systemic risk of an asset, indicating its sensitivity relative to the overall market [7, 8]. Monte Carlo simulation, which simulates a myriad of potential price paths, forecasts the future distribution of assets and provides investors with crucial risk measures such as value-at-risk (VaR) and conditional value-at-risk (CVaR) [9, 10]. These measures are particularly significant in carbon markets, where they provide a more comprehensive risk estimation than volatility and beta. VaR stands out for its ability to offer a probabilistic estimate of the maximum potential loss within a specific confidence interval, thus identifying risk more precisely than the general fluctuation trends indicated by volatility. Building on VaR, CVaR quantifies the expected loss in the worst-case scenarios, thereby capturing extreme risks that are often neglected by volatility and beta. The focus of both the VaR and CVaR on potential extreme losses renders them indispensable for obtaining detailed and accurate risk profiles in the carbon market. They effectively address the limitations of traditional measures by encompassing a broader spectrum of risks, particularly in markets characterised by non-linear and asymmetric dynamics [11, 12]. However, traditional methods for estimating VaR and CVaR, such as Monte Carlo simulations require many simulations to achieve a reliable estimate and often operate on implicit assumptions of data probability distributions, which may only occasionally reflect reality [13]. This has led to the exploration of quantum algorithms as promising alternatives. Notably, studies show that quantum algorithms can calculate risk contributions in credit portfolios more efficiently by employing quantum oracles and algorithms for the simultaneous estimation of multiple random variables that are crucial for CVaR contributions [14]. Furthermore, a comprehensive review of the field of quantum computing in finance demonstrates the significant computational benefits that modern approaches offer over traditional techniques [15]. Considering these challenges, the exploration of quantum algorithms has emerged as a promising frontier, providing a more efficient and precise avenue for risk estimation.

Existing quantum computing research on risk estimation revolves around the simulation process and utilises quantum amplitude estimation (QAE) to improve computational efficiency and accuracy [16, 17]. However, most studies have simulated distributions using pre-assumed models to generate data, such as the Gaussian conditionally independent model [18, 19]. Such assumptions may not align with market dynamics and may cause bias. Conditional generative adversarial networks (CGANs) are widely used to model data distributions, and their quantum version, the quantum conditional generative adversarial network (QCGAN), is regarded as a technology that holds great potential for data distribution simulation in real-world applications such as image generation [20–22]. By combining QAE, QCGAN, and binary search, our research pioneers the application of quantum computing in carbon market risk estimation. The integration of the QCGAN significantly improves efficiency and accuracy, whereas utilising a binary search solves the problem of the QAE's inefficiency in searching for VaR, thereby improving the efficiency of the entire risk estimation. Collectively, these improvements represent significant advancements in risk estimation.

To solve the above-mentioned problems, a quantum computing-based framework is proposed to estimate risk in the carbon market. The contributions of this study are summarised as follows:

(1) The development of the quantum conditional generative adversarial network-quantum amplitude estimation (QCGAN-QAE) framework represents a pioneering effort to apply quantum computing to carbon market risk estimation. This framework is unique in its combination of the QCGAN and QAE, offering enhanced accuracy and efficiency in market risk estimation. This novel approach positions this study at the forefront of integrating quantum computing techniques into financial risk analysis. To the best of our knowledge, this study is among the first to apply quantum computing to estimate carbon market risk.
(2) An innovative redesign of the quantum generator circuit within the QCGAN has been achieved. This restructuring involves reordering the data entanglement and rotation layers, supplemented by the introduction of a quantum fully connected layer (QFC) before the rotation operations. This advancement in quantum circuit design is crucial for improving the capacity of the model to process complex financial data, resulting in more precise risk estimation.
(3) To improve the computational efficiency, the binary search approach is incorporated with QAE, rooted in quantum coding, which adeptly delineates the spatial boundaries for good states in QAE.

The remainder of this paper is organised as follows: Section 2 introduces related studies on the QCGAN and QAE. The proposed QCGAN-QAE framework for carbon market risk estimation and modified QCGAN and QAE methods are presented in Section 3. Section 4 presents the experimental results of the proposed methods using the EU ETS data. Finally, the conclusions and discussion are presented in the last section.

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2 | BACKGROUND

Machine learning algorithms are widely used in the energy field [23, 24]. Quantum computing leverages the principles of quantum mechanics to process information in fundamentally different methods than classical computers [25]. For carbon market risk estimation problems, the QCGAN generates a risk distribution. The QAE is used to efficiently measure the distribution and obtain the risk estimation indicators, VaR and CVaR [26].

To deepen our understanding of the quantum computational principles that underpin the proposed methodology, it is imperative to introduce the key concepts of quantum computing. At the core of quantum computing, quantum states are represented in complex vector space. These states are denoted using the ket notation $|\cdot\rangle$, such as $|\psi\rangle$ for a quantum state ψ. The corresponding bra notation $\langle\cdot|$, exemplified by $\langle\psi|$, signifies the complex conjugate transpose of the ket. These notations, which are integral to the Dirac notation, are essential for describing the quantum states and operations that form the basis of the proposed QCGAN and QAE approaches.

2.1 | Quantum conditional generative adversarial network

CGANs incorporate conditional data to boost generation accuracy [27, 28]. Quantum properties, specifically superposition and entanglement, improve the performance of generative modelling [22]. The key components of a QCGAN include a generator, discriminator, and an objective function.

2.1.1 | Generator

The generator in the QCGAN is constructed as a parameterised quantum circuit (PQC), which is a quantum circuit with adjustable parameters that can be optimised during training. The PQC encodes conditional constraints as an input quantum state $|\psi(y)\rangle$ and outputs a quantum state $|\psi_{\text{out}}\rangle$ as the generated data sample. For a specified condition state $|\psi(y)\rangle$ and an initialised random state $|\psi(z)\rangle$, the quantum transformation governed by the PQC is given by:

$$|\psi_{\text{out}}\rangle = U(\theta)(|\psi(y)\rangle \otimes |\psi(z)\rangle), \tag{1}$$

where $U(\theta)$ denotes the PQC with parameters θ, and $\otimes$ represents the tensor product.

2.1.2 | Discriminator

The discriminator D in the QCGAN integrates a classical neural network to classify quantum states into real or generated data samples. For input data χ, D yields a probability p indicating the authenticity of χ, to represent whether it is genuine or generated by the generator. This is formally expressed as

$$p_{\text{QCGAN}} = D(\chi) \tag{2}$$

The primary training goal for D is to accurately classify the quantum states.

2.1.3 | Objective function

The key to the CGAN is adversarial training, which minimises the quantum Jensen–Shannon divergence. During training, discriminator D attempts to maximise the differentiation of the real data from the output of generator $G(z|y)$. Simultaneously, generator G aims to generate data that the discriminator considers real, that is, to minimise the difference between the generated data and real data. The optimisation of the QCGAN can be formulated as a min-max problem:

$$\min_G \max_D (E[\log D(x|y)] + E[\log(1 - D(G(z|y)))]), \tag{3}$$

where $E[.]$ represents mathematical expectation.

2.2 | Quantum amplitude estimation

QAE aims to estimate the amplitude of a particular state within a quantum system [29]. This algorithm can be regarded as a natural extension of quantum amplitude amplification (QAA) [30].

In quantum mechanics, amplitude directly corresponds to probability. Consider the quantum state [31]

$$|\psi\rangle = \sqrt{1 - p_{\text{QAE}}}|\psi_0\rangle + \sqrt{p_{\text{QAE}}}|\psi_1\rangle, \tag{4}$$

where $|\psi_0\rangle$ and $|\psi_1\rangle$ are normalised states, and p_{QAE} is a real number. The goal of the QAE is to estimate the amplitude a and the square of the amplitude a is equivalent to the probability, that is $|a|^2 = p_{\text{QAE}}$. Within the QAE framework, the angle θ is used to represent the amplitude a, with $a = \sin\theta$, leading to $\sin^2\theta = p_{\text{QAE}}$. This transformation provides an avenue for the use of rotation operations to modulate the amplitude. The formula after the substitution is as follows:

$$|\psi\rangle = \cos\theta|\psi_0\rangle + \sin\theta|\psi_1\rangle. \tag{5}$$

Based on the above analysis, the measurement of probability is converted into the measurement of angle θ, and the core is to amplify θ. To amplify θ, the key is to have an $\mathcal{A}$ that can be used repeatedly and an oracle $\widehat{\chi_{\varphi_0}}$ that can identify the marked state $\widehat{\chi_0}$. Therefore, we consider the following operators:

$$\widehat{\chi_\varphi} = I - 2|\varphi\rangle\langle\varphi| \tag{6}$$

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This operator assigns a sign flip to the state $|\varphi\rangle$. Given an oracle $\widehat{\chi_{\varphi_0}}$ that interacts specifically with the marked state $|\psi_0\rangle$, and with the ability to repeatedly apply $\mathcal{A}$ and $\mathcal{A}^\dagger$, the operation $\mathcal{A}\widehat{\chi_0}\mathcal{A}^\dagger = I - 2|\psi\rangle\langle\psi|$ can be executed on the encoded state $|\psi\rangle$. Thus, the "Grover" operator $\mathcal{Q}$ is defined as [29]:

$$\mathcal{Q} = \mathcal{A}\widehat{\chi_0}\mathcal{A}^\dagger\widehat{\chi_{\varphi_0}}. \tag{7}$$

This operator ensures that the system reaches a certain state after a certain number of Grover iterations.

$$\psi_E(k) = \mathcal{Q}^k|\psi\rangle = \cos((2k+1)\theta)|\psi_0\rangle + \sin((2k+1)\theta)|\psi_1\rangle \tag{8}$$

By appropriately selecting the iteration count k, the amplitude of the target quantum state can be amplified to the desired level, facilitating accurate amplitude estimation during the measurement.

2.3 | Market risk evaluation metrics

VaR and CVaR are risk metrics commonly used in finance to quantify potential losses in the value of a risky portfolio over a given period at a given confidence level. VaR estimates the maximum potential loss of an investment portfolio [32]. The VaR formula is as follows:

$$P(L > \mathrm{VaR}_\alpha) = 1 - \alpha, \tag{9}$$

where P represents the probability, L represents the loss over a given period, and α represents the confidence level.

The CVaR calculates the expected value of the loss because it already exceeds the VaR, which makes it better to consider extreme losses [33]. The CVaR is calculated as follows:

$$\mathrm{CVaR}_\alpha = \frac{1}{1-\alpha}\int_{L>\mathrm{VaR}_\alpha} Lf(L)\,dL, \tag{10}$$

where $f(L)$ represents the probability density function of the loss L.

3 | QUANTUM-BASED CARBON MARKET RISK ESTIMATION FRAMEWORK

3.1 | Carbon market risk estimation based on quantum algorithm

To effectively address the challenges faced in previous research, including requiring a large number of measurements, consuming computing resources, and having implicit assumptions that reduce accuracy, a quantum computational framework based on the QCGAN and QAE algorithms is introduced. The synergy of these two algorithms enables us to refine the risk estimates for carbon markets. The overall structure of the proposed framework is shown in Figure 1 and includes data preprocessing, risk distribution learning via QCGAN, risk measurement through QAE, and risk evaluation. Initially, data preprocessing converts the raw market data into standardised quantum state data. Subsequently, in risk distribution learning, the QCGAN is trained with quantum state data to generate simulated one-day risk distributions. Risk distribution measurement combines the QAE with the binary search method to measure the previously obtained risk distribution, determine the target quantum state corresponding to VaR, and measure the probability distribution of the quantum state required to calculate CVaR. Finally, VaR and CVaR are decoded and calculated using the measured results in the risk evaluation section. The specific details of the framework given below.

3.1.1 | Data preprocessing

At this stage, the loss rate is calculated and converted into binary quantum encoding to serve as the input, providing the foundation for subsequent quantum computations. We calculated the loss rate $[y_1, \dots, y_T]$, which is the negative value of the rate of return R, from the carbon market price data $[p_0, \dots, p_T]$, T indicates the total number of days. The formula used is as follows:

$$y_t = -R_t = -\frac{p_t - p_{t-1}}{p_{t-1}}. \tag{11}$$

The historical loss rate for the past N days $[y_{t-N}, \dots, y_{t-1}]$ is used as conditional data to simulate future data generation for the object. The future loss rate y_t is used as the object data χ_t. The loss rate is chosen to measure risk rather than price because, starting from the definitions of VaR and CVaR, investors tend to pay more attention to the rate of change in price than to its absolute value. Binary encoding techniques are used to convert classical data into quantum data. The loss rate was first normalised using the min–max method and evenly divided into 2^n segments according to the numerical value, where n indicates the number of qubits required for encoding. This normalisation process is crucial for mitigating potential biases in the historical data. By scaling the data within a uniform range, we reduced the influence of outliers and ensured that each data point contributed proportionately to model training. Subsequently, we converted the numerical value of each segment into a binary format for processing in quantum computing. To train and evaluate the model, we divided the dataset into two key parts: a training set and a test set. The training set is used to train the model to understand and capture data characteristics. The test set is used to evaluate the performance of the model.

3.1.2 | Risk distributions learning

Subsequently, we deployed the QCGAN to simulate the potential future loss rate distributions. We provide a quantum generator with a historical loss rate previously transformed into

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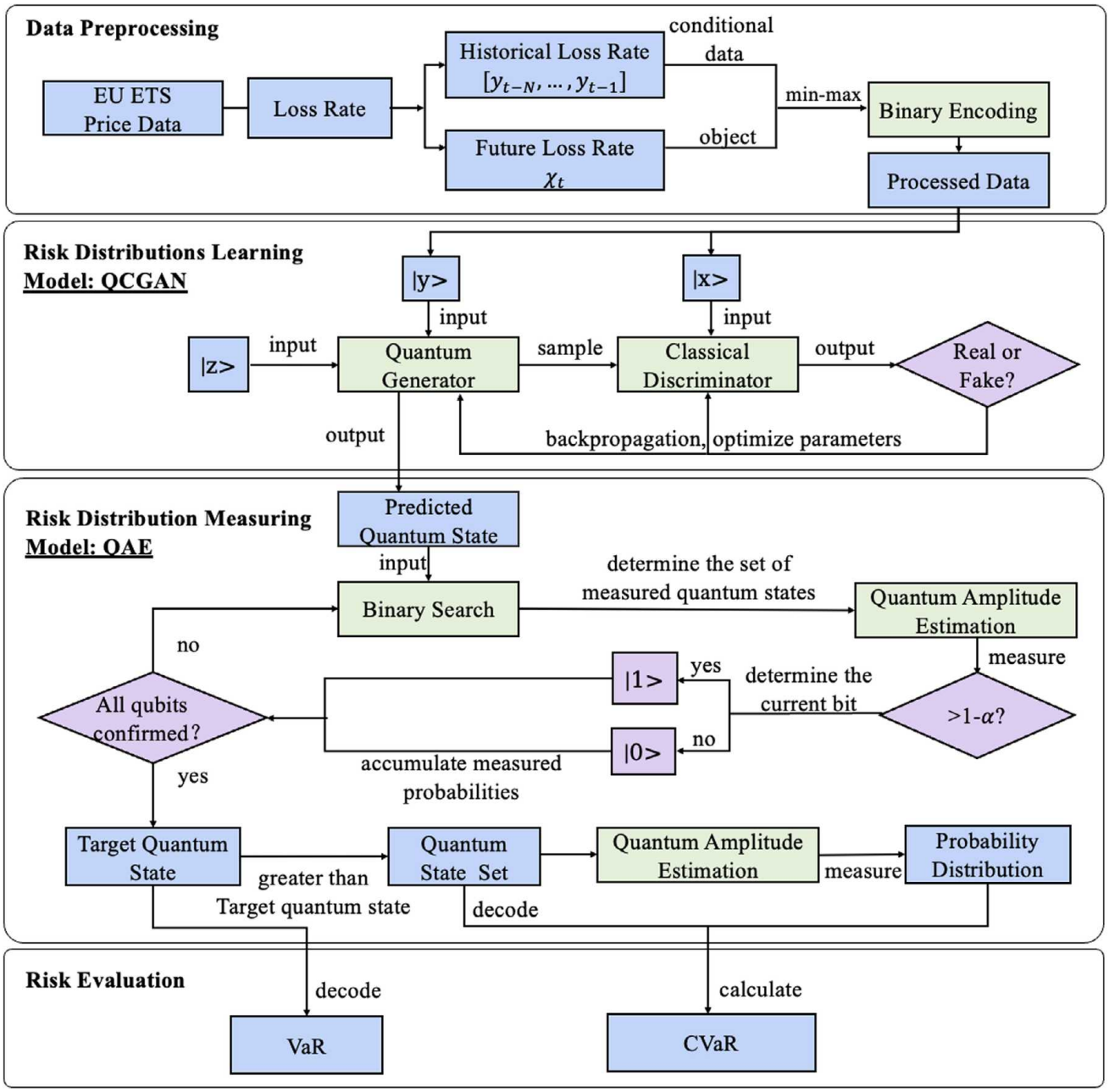

FIGURE 1 Carbon market risk estimation based on QCGAN-QAE framework. CVaR, conditional value at risk; EU ETS, European Union Emissions Trading System; QAE, quantum amplitude estimation; QCGAN, quantum conditional generative adversarial network; VaR, value at risk.

quantum states as conditional data. A special feature of the QCGAN is the incorporation of conditional data $|y\rangle$ and noise $|z\rangle$. Among them, y indicates the previously preprocessed quantum data of the past n days, that is, $[y_n, y_{n-1}, \cdots, y_{t-1}]$; z denotes the quantum noise. This noise provides randomness to the generator and serves as supplementary information to assist the generator in generating samples with distinct characteristics. Consequently, the generator combines these conditional data with the noise to produce samples that represent potential future loss rate distributions. The samples generated by the generator are passed through a classical discriminator. The most important function of the discriminator is to differentiate between the samples created by the generator and the actual data samples, thereby providing an evaluation score in the process. Upon receiving feedback from the classical discriminator, both the quantum generator and discriminator optimise their parameters using a backpropagation algorithm. With each iteration, both the generator and discriminator parameters are progressively optimised. Once a predetermined number of iterations is reached or a specific accuracy threshold is met, the generator simulates future loss-rate distributions with extremely high accuracy.

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The output of the QCGAN, the predicted quantum state, is a complex quantum state that encodes the probability distribution of the potential future loss rates in the market. This state is represented by a superposition of multiple quantum-basis states, each corresponding to a specific market scenario. Each quantum basis state is a unique combination of qubits that represents a possible outcome in the market. For instance, the basis state in the QCGAN represents a particular loss rate in the market. The probability amplitude associated with each state quantifies the likelihood of a market scenario occurring.

3.1.3 | Risk distribution measuring

To improve the efficiency of measuring the predicted quantum state generated by the quantum generator, we obtained the target quantum state corresponding to VaR, and the quantum state and its probability required to calculate CVaR using QAE combined with a binary search. The process starts with a binary search that divides the quantum state into two subspaces based on the median value of all quantum states, that is, $|10 \ldots 0\rangle$. A set of quantum states composed of quantum states greater than or equal to the median is measured, that is, $|1x \ldots x\rangle$. We then use QAE to measure the probability sum of the quantum state set quantum states. The next step is to confirm the target quantum state. Specifically, the probability sum of measured quantum states is compared with $1 - \alpha$, where α denotes the confidence level. If the probability sum is greater than $1 - \alpha$, which means that the target quantum state is larger than the median, then the current qubit is determined to be $|1\rangle$; otherwise, the target quantum state is smaller than the median, the current qubit is $|0\rangle$, and the probability sum of measured quantum states is accumulated. Thus, the current qubit of the target quantum state is confirmed. Return to the binary search to search for a more precise target quantum state and determine the next qubit until all qubits are determined. Finally, obtain the entire target quantum state, which is the quantum state corresponding to VaR. Upon determining the target quantum state, all quantum states that are greater than the target quantum state correspond to extreme loss rates greater than the VaR. This set of quantum states is required to calculate the CVaR. The QAE is used to measure these quantum states and obtain their probability distribution, which provides a data source for subsequent calculations.

3.1.4 | Risk evaluation

The final step is risk evaluation. For VaR, initially, the target quantum state is determined through QAE combined with binary search, VaR can be obtained just by decoding the target quantum state. For CVaR, the CVaR is calculated by decoding the quantum states in the quantum state set and the corresponding probabilities. Given the discrete framework of quantum computation, the CVaR computation transitions from the conventional integral to a discrete summation, which simplifies the expression and aligns with the operational logic of quantum algorithms. The specific formula is as follows:

$$\mathrm{CVaR}_\alpha = \frac{1}{1-\alpha} \sum_{L > \mathrm{VaR}_\alpha} L f(L), \tag{12}$$

where α is the confidence level, L indicates the loss rate obtained after decoding the quantum state set, $f(L)$ represents the corresponding probability.

3.2 | Improved quantum conditional generative adversarial network for generating simulated risk distribution

Accurately simulating the distribution of the loss rate is crucial for risk estimation, and the QCGAN was employed and improved upon in this study. Specifically, the PQC structure of the generator is enhanced. Compared to previous models in quantum-based risk estimation, the improved QCGAN exhibits better performance in generating precise quantum states. The generator, discriminator, and learning process of the improved QCGAN for risk distribution generation are described as follows.

The primary component of the generator is the PQC, which simulates the fundamental dynamics of the loss rate. The PQC includes several parameterised quantum gates, such as rotation gates and controlled-not (CNOT) gates. The parameters of these gates are optimised through an iterative process based on the feedback from the discriminator. The proposed circuit comprises three stages: preparation, entanglement, and rotation. The preparation stage encodes the market condition state $|y\rangle$ as a quantum state. The entanglement stage ensured that the encoded condition state interacted well with the noise. The rotation stage is responsible for generating the risk distribution. The design and operation of these three modules are described in this section. To enhance the expressive capability of the generator and capture complex relationships within the market data, we introduce a QFC in the entanglement stage. The QFC layer comprises a series of CNOT gates and rotation gates to achieve full connectivity among qubits, which can facilitate interactions among all qubits. This fully connected structure ensures that no crucial interaction information is lost during the entanglement process, thereby enhancing the accuracy of carbon market simulation. The circuit structure is illustrated in Figure 2. The details of the QFC are introduced in this section.

For the discriminator, a classic fully connected neural network is employed to distinguish between real and generated data as a binary classification problem [34]. The inputs of the discriminator are randomly chosen from the real data $x|y$ or quantum generator $G(z|y)$. The output of the discriminator is scalar in the range [0,1], representing the probability that the input data are regarded as real data. The neural network comprised multiple convolution layers, each followed by a rectified linear unit (ReLU) activation function, with the final layer utilising a sigmoid activation function to map the output to the range (0,1). The dimensions of the input layer in the

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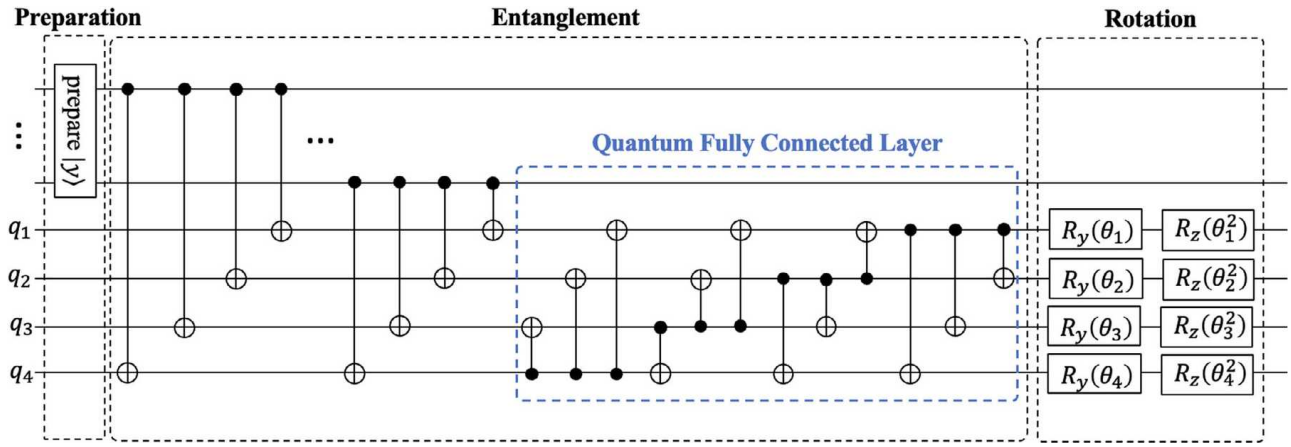

FIGURE 2 Quantum circuit of the generator (using the generation of data that requires four qubits as an example).

discriminator match the number of qubits in the quantum generator to facilitate the direct interaction between quantum and classical data [35]. The structure of the hidden layers can be adjusted according to the task complexity.

For the learning process, both the generator and discriminator are co-trained using the Adam algorithm. The discriminator training frequency was adjusted for stability, and binary cross-entropy loss was used for optimisation. Throughout the training process, the Jensen–Shannon divergence s used to measure the difference between the distribution of quantum states generated by the generator and the real data distribution. Based on these measurements, adjustments are made to refine the generator. When optimising the parameters, the traditional neural-network-based discriminator can directly compute the gradient of the loss function through backpropagation. However, in variational quantum circuits, owing to their complex quantum properties, we cannot directly calculate the gradient of the loss function as in traditional neural networks. Therefore, we use the parameter shift rule to estimate the partial derivatives [35], thereby enabling the use of the same quantum circuit to compute gradient information. This collaborative optimisation process of the generator and discriminator enables the QCGAN to generate loss-rate distributions accurately. Through continuous feedback and optimisation loops, the generator can produce outputs that are increasingly closer to the actual data distribution, whereas the discriminator's distinguishing ability gradually improves, thereby achieving good results in simulating the loss-rate distributions.

The improved QCGAN circuit structure plays a crucial role in enhancing the accuracy of simulating loss-rate distributions. By reordering the data entanglement and data rotation layers and incorporating QFC, we significantly improved the circuit's ability to capture complex correlations within market data. The specific details of our generator modifications are as follows.

3.2.1 | Swap entanglement and rotation

A novel generator quantum circuit is proposed to accurately simulate the quantum state of the loss-rate distribution. As illustrated in Figure 2, the quantum circuit of the generator comprises three key stages: preparation, entanglement, and rotation. In earlier studies, quantum circuits typically proceeded through the rotation stage before the entanglement stage [22]. However, we discovered that placing the entanglement stage before the rotation stage significantly enhanced the efficiency and expressive capabilities of the quantum circuit. First, through the prior entanglement operation, strong quantum correlations were established before the rotation operation, providing favourable conditions for the subsequent rotation stage. Such strong quantum correlations enable the quantum circuit to better explore and utilise the multi-dimensional characteristics of the quantum space during the rotation stage, thus improving its expressive capability of the quantum circuit. Second, based on the already entangled quantum states, the rotation operations can adjust the quantum states more accurately and purposefully, reducing the number of required rotation operations, thereby improving the efficiency and accuracy of the quantum circuit. The specific functions of each stage are described in detail below.

In the preparation stage, the main goal of quantum circuits is to encode binary quantum data in quantum circuits. This was accomplished by passing it through an X-gate. For example, encoding $|0101\rangle$ applies X gates to qubits where $|0\rangle$ is located, that is, the first and third qubits. Such operations not only ensure that quantum states start from a representation consistent with classical conditions but also lay the foundation for subsequent quantum interactions. The X gate flips the state of a qubit from $|0\rangle$ to $|1\rangle$ and vice versa, aligning the initial quantum state with the classical input data.

The subsequent stage is entanglement, which significantly deviates from conventional architectures. Conventionally, the rotational stage precedes the entanglement stage. Our revised model prioritises the establishment of a coherent relationship between the conditions and noise qubits ahead of rotation. Conditional qubits refer to qubits that encode conditional data, whereas noise qubits refer to qubits that encode noise. CNOT gates play a pivotal role in creating deterministic entanglements between these qubit sets. A CNOT gate entangles two qubits such that the state of one (control qubit) determines the

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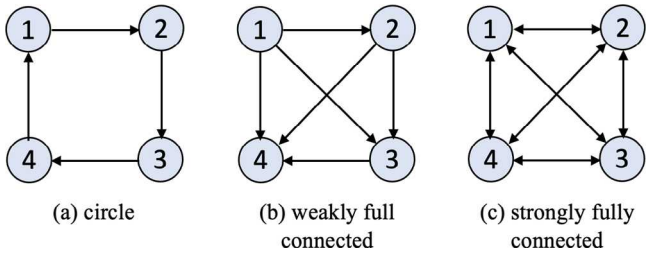

FIGURE 3 Three common qubit-to-qubit connection topologies.

outcome of the other (target qubit). This entanglement ensures that the quantum states of the noise qubits are affected by the condition, thereby improving the efficiency and accuracy of the model's simulation capabilities.

The last stage is rotation, where qubits undergo a sequence of operations using RY and RZ rotation gates [36]. These rotation gates are parameterised by values in θ, enabling continuous adjustment to better simulate the desired quantum data states. RY and RZ gates are crucial for adjusting the probability amplitudes and phases of the quantum states, respectively. By fine-tuning the parameters of these gates, we can control the evolution of the quantum states to align them with the specific requirements of our risk estimation model.

By deploying these rotation and entanglement gates in a structured and orderly manner, the quantum states evolve coherently and progressively converge to the target configuration.

3.2.2 | Quantum fully connected layer

A fully connected layer in classical neural networks suggests that each neuron in the previous layer is connected to every other neuron in the subsequent layer. Similarly, in the QFC layer, every qubit interacts with another qubit, facilitating a full exchange of quantum information across the circuit. This interaction is achieved by applying a CNOT gate between every pair of qubits, ensuring that each qubit influences every other.

The introduction of the QFC layer into the QCGAN-QAE model significantly bolsters its ability to accurately simulate complex market dynamics. By fostering interactions among all the qubits, the QFC layer ensures comprehensive information processing within the quantum circuit, thereby enhancing the framework's capacity to represent and learn from multifaceted market data. This holistic approach to information flow in our quantum model not only accelerates convergence during the training phase but also contributes to the refined representation of quantum states, which is crucial for accurate risk estimations.

As shown in Figure 3, considering a four-qubit system as an example, the common connection topologies of qubits include three types: circular, weakly fully connected, and strongly fully connected. Previous research has shown that although a circular topology requires fewer quantum gates, it requires more layers to satisfy the entanglement requirements of complex tasks [35–38]. The fully connected layers currently used in QGAN are weakly fully connected (see Figure 3b); that is, only a one-direction connection exists between all qubits, but this connection is not sufficient and may reduce the learning ability [22, 39]. Therefore, this study applies a strongly connected and fully connected layer to the entanglement stage [40]. The specific connection method and location of the QFC in the generator are depicted in Figure 3.

3.3 | Quantum amplitude estimation combined with binary search for measuring risk distribution

In our quantum-based risk estimation method, the QAE is central for accurately and efficiently estimating the risks associated with the market. The QAE measures the quantum states generated by the QCGAN. In contrast to classical measurements, the QAE leverages the power of quantum computing to efficiently measure small probability events through amplitude amplification. Using QAE, we transcend the limitations of traditional measurement methods, which often ignore rare events by only sampling high-probability events when computational resources limit the sample size. In the context of carbon market risk estimation, tail events can have significant consequences and the ability to capture and estimate these rare events is necessary. In contrast to its application in chemistry, QAE is mainly used to determine the ground-state energy of molecules [41]. When applied to risk estimation, QAE is tasked with precisely measuring the probability distribution and computing the cumulative distribution function (CDF) rather than solely focusing on a single peak value or the optimal solution. In our study, the CDF refers to the sum of the probabilities that the loss rate is greater than the specified loss rate. However, existing QAE methods for estimating the CDF must traverse all probability distributions. This exhaustive traversal is essential for accurately pinpointing the ROIs to establish a full quantum space. This approach requires many measurements, thereby reducing the overall model efficiency.

To enhance the efficiency of CDF estimation, a binary search was integrated with the QAE. A binary search searches for the target quantum state corresponding to a certain confidence level and individually determines the qubits of the target quantum state. The space was divided into 2^n parts. By leveraging the characteristics of binary encoding, we pinpoint the quantum state starting from the highest qubit, and each search reduces the space size to which the target quantum state belongs by half and determines the qubit of the current bit. Importantly, this approach significantly reduces the time complexity of VaR. The time complexity is logarithmic, specifically $O(\log N)$, where $N = 2^n$ is the total number of possible states. This logarithmic time complexity is a result of the binary search algorithm, in which each iteration halves the search space, enabling us to locate the VaR result within no more than n iterations, where n indicates the number of qubits required to encode the quantum data.

Moreover, the scalability of the proposed framework with respect to the number of qubits is a critical consideration. As the number of qubits increases, the framework can represent

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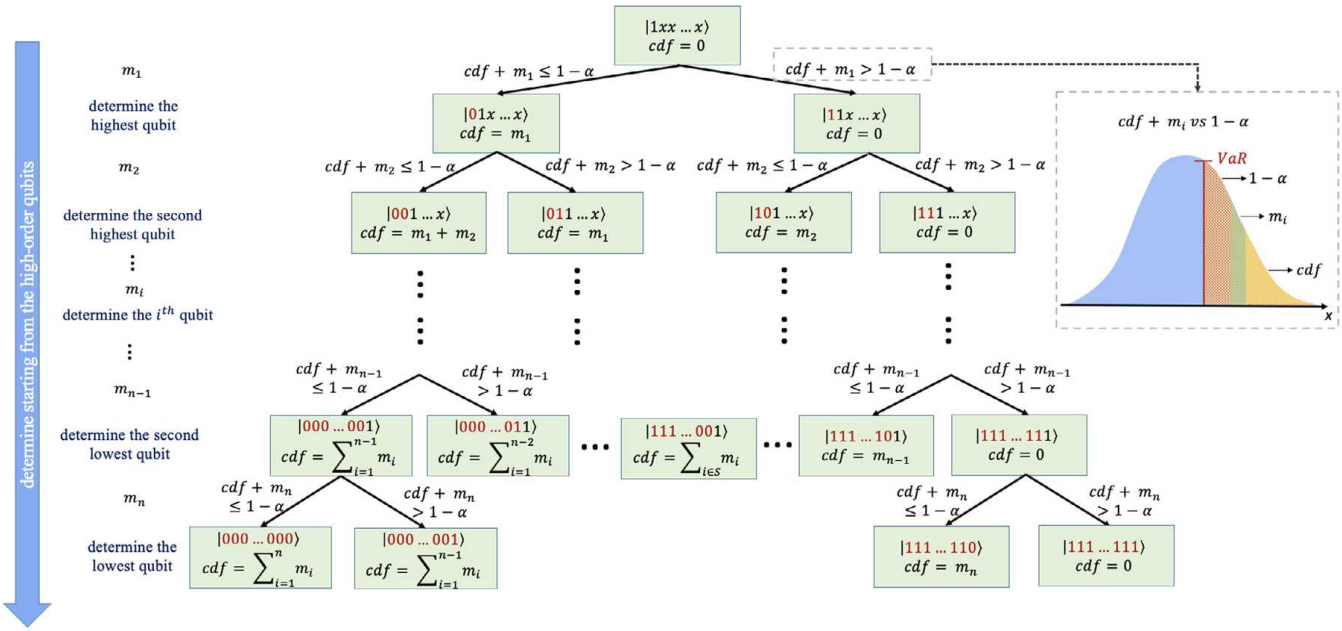

FIGURE 4 Quantum amplitude estimation combined with binary search. cdf, cumulative distribution function; VaR, value at risk.

and process more complex high-dimensional data. However, this increases the computational resources required for accurate state preparation and measurement. Our framework is designed to balance increased representational power with computational and practical constraints. Although scalability is subject to current quantum computing capabilities, our model remains efficient and feasible within a realistic quantum computing context, demonstrating its potential scalability as quantum technology advances.

Therefore, addressing the problem of encoding continuity is crucial. The continuity of quantum coding refers to the adjacency of numerical ranges that correspond to the quantum state. The discontinuity of encoding indicates the discontinuity and inaccuracy of the estimated CDF, which in turn leads to the inaccuracy of the estimated VaR and CVaR. Confirming qubits starting from the highest bit ensures that all uncertain qubits appear in the lower bits so that all sets of quantum states to be measured are continuous. For example, the first two digits of $|10x\rangle$ are determined and the quantum sets to be measured are $|100\rangle$ and $|101\rangle$. The numerical intervals corresponding to these quantum states are continuous. However, if we do not start from the highest bit, for example, if we try to locate the state $|1x1\rangle$, then states $|101\rangle$ and $|111\rangle$ would be identified, leading to a discontinuity in the corresponding classical value because $|110\rangle$ is missing. This continuity lends rationality to a set of target quantum states.

The details of the QAE combined with a binary search are as follows: The framework is illustrated in Figure 4.

In m1, the binary search method is used to divide the quantum state space into two parts with the median of the quantum state as the boundary: one part is $|1x\ldots x\rangle$ and the other part is $|0x\ldots x\rangle$. Among them, what is measured is the space of quantum states greater than the median, that is, $|1x\ldots x\rangle$. That is, the median is the midpoint of the x-axis in the probability distribution diagram. This method involves determining the location of the "good" space relative to the median, so as to find the "good" space. Notably, with a binary search, our attention remains focused on the determined and subsequent quantum positions, whereas distant positions are ignored. For example, in a four-qubit system, having determined the first two positions as $|10\rangle$, we would mark $|101x\rangle$ (the final '1' that arises owing to a binary search).

Subsequently, the quantum states were measured and validated. Comparing the measured probability m_i of the current set of quantum states determined by the binary search method with the sum of the previous cumulative probabilities cdf against a default confidence level α.cdf refers to the sum of probabilities greater than or equal to the current loss rate. If $(cdf + m_i)$ exceeds $(1-\alpha)$, which means that the "good" space is larger than the median, the current position is confirmed as $|1\rangle$; otherwise, means that the "good" space is smaller than the median, the current position is confirmed as $|0\rangle$. For positions determined to be $|0\rangle$, we must document the combined probability of the marked states, that is $cdf_{\text{new}} = cdf_{\text{original}} + m_i$. Using this method, a good space is determined, and the size is reduced by half. Revisiting the previous example in a four-qubit system, the quantum state corresponding to the VaR determines that the first two qubits are $|10\rangle$. This means that in the first previous comparison, the initial cdf is zero, and the comparison found that the sum $(cdf + m_1)$ is the probability sum of $|1xxx\rangle$, that

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is, m_1, is greater than $1-\alpha$, so the first qubit is confirmed to be $|1\rangle$, and not store m_1, so the $cdf_{\text{new}} = cdf = 0$. In the second comparison, because the cdf is 0. The comparison found that the sum $(cdf + m_2)$ is the probability sum of $|11xx\rangle$, that is, m_2, was smaller than $1-\alpha$, so the second qubit was confirmed to be $|0\rangle$, and stores m_2, so the $cdf_{\text{new}} = cdf + m_2 = m_2$, that is, the probability sum of $|1100\rangle$, $|1101\rangle$, $|1110\rangle$, and $|1111\rangle$.

The median is then determined again based on the binary search method, and the above steps are repeated until all qubits of the quantum state are determined or the range is determined. This method checks the aggregated probabilities of confirmed states $(cdf + m_i)$ against $(1-\alpha)$, ensuring that by the iteration's end, we possess a state closest to the desired configuration, the quantum state corresponding to VaR. According to the quantum state corresponding to VaR, we find the quantum state whose loss rate is greater than VaR, measure its probability, and finally calculate CVaR.

4 | CASE STUDY

4.1 | Settings

4.1.1 | Data settings

As the largest participant in the EU ETS, the European Climate Exchange (ECX) is a critical indicator of the global carbon market. Our analysis focuses on ECX price data, providing insights into the market's volatility and trend behaviours, which are crucial for understanding the accuracy and efficiency of our proposed QCGAN-QAE model in a real-world context. For this study, we use the daily carbon price from the EU ETS market to calculate the daily rate of return, and then take a negative number for the daily return to obtain the loss rate. The data are obtained from the European Energy Exchange and are expressed in euros per ton. Given the phased nature of the EU carbon market, we focus on the third phase (2013–2020) of market development. The dataset employed in this study includes carbon market prices from 1 January 2015 to 31 December 2020. This period was carefully selected to provide a comprehensive view of market behaviour. The dataset, comprising 1544 data points, excludes public holidays to maintain consistency. It was strategically divided into two subsets, an 80% training set and a 20% test set, without shuffling, to ensure the integrity of the time-series data for accurate risk estimation. To enhance coding efficiency and accuracy, as well as to accelerate and stabilise data convergence, all historical loss rates processed from price data were min-max normalised into the range of $[-1, 1]$ and then evenly divided into 2^8 segments. Finally, we converted the numerical value of each segment into a binary format for processing in quantum computing.

4.1.2 | Experimental settings

The study was conducted using Python version 3.8.10. We utilised Qiskit 0.20.0, IBM's open-source software development kit for quantum computing. Qiskit facilitates the construction and simulation of the quantum circuits central to the QCGAN-QAE model. For data manipulation and preprocessing, we employed Pandas 1.3.5 and NumPy 1.23.5. Matplotlib 3.5.2 was used for all data visualisation needs, ensuring clear and informative graphical representations of our data and results. The classical component of our model, particularly the discriminator within the CGAN, was implemented using Scikit-learn 1.0.2. The COBYLA optimiser from Qiskit's Aqua Components was used to optimise the quantum circuits.

4.1.3 | Criteria for VaR/CVaR

The evaluation criterion for VaR/CVaR was back testing. The backtesting of VaR in this study was performed using the Kupiec test. This test examines whether the number of times the actual loss rates exceed the VaR estimates follows a binomial distribution with a success probability equal to the VaR confidence level. Let T be the total number of observations, N as the number of failures (i.e., the number of times the actual loss rate exceeds the predicted VaR), α as the given confidence level and p is the theoretical failure ratio, that is, $1-\alpha$. The failure ratio (r) and likelihood ratio (LR) [42] can then be expressed as

$$r = N/T \tag{13}$$

$$LR = 2\ln\left(\frac{(1-r)^{T-N}(r)^{N}}{(1-p)^{T-N}(p)^{N}}\right) \tag{14}$$

To facilitate a direct comparison between the observed failure ratio and the expected probability, we define the discrepancy measure Δr. This measure, Δr, is the absolute difference between the failure ratio, r, and the expected probability, p, mathematically expressed as $\Delta r = |r - p|$. The smaller the value, the more accurate the estimation. If LR surpasses the critical value (a chi-squared distribution with one degree of freedom) at a given significance level, the VaR model is rejected. Among these, the Kupiec test for the LR is the most basic and important. Only after passing the Kupiec test can the comparison of Δr be meaningful.

The backtest of CVaR is the difference between the losses and CVaR (DLC) tests. The DLC back test focuses on the difference between the actual loss rates that exceed the VaR and CVaR values [43]. This statistic is defined as

$$\text{DLC} = \left|\frac{1}{N}\sum_{i=1}^{N} X_i - \frac{1}{N}\sum_{i=1}^{N}\text{CVaR}_i\right|, \tag{15}$$

where X_i indicates the actual loss rates that exceeded VaR. The smaller the DLC, the closer the expected value of the CVaR is to the expected value of the actual loss rates, and the higher the accuracy of the CVaR measurement.

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4.1.4 | Quantum computer computational time cost

Our primary goal is to ascertain the real-world potential of quantum hardware in the context of risk assessment of carbon market returns. Therefore, it is crucial to evaluate the actual computational time that QAE would require on a quantum processor, not just on classical computers [42].

Therefore, we consider both the time required to simulate the distribution using the QCGAN quantum generator and the subsequent measurements required for the QAE process. Each quantum process is associated with a distinct circuit depth: $D_{\mathrm{QCGAN},i}$ for QCGAN and $D_{\mathrm{QAE},i}$ for QAE. The unified computation time, considering a single execution of these two processes, is calculated as follows [44]:

$$T_i = T_P + T_G \times \left(D_{\mathrm{QCGAN},i} + D_{\mathrm{QAE},i}\right) + T_M, \tag{16}$$

where T_P indicates the time required to prepare the initial state; T_G indicates the average duration of the quantum gate; and T_M indicates the time required to measure the qubits.

Extending to m distributions in a scenario requires simulations and measurements with a cumulative computational time of

$$T_{\mathrm{total}} = \sum_{i=1}^{m} T_i \tag{17}$$

For this estimation, we adopt the following time scales, which reflect aggressive yet realistic predictions for state-of-the-art quantum devices: $T_P + T_M = 1\ \mu\mathrm{s}$, $T_G = 10$ ns.

4.1.5 | Convergence criteria in CGAN and QCGAN models

The convergence of the CGAN and QCGAN is fundamentally assessed by stabilising the loss functions, including the generator and discriminator losses. As the training epochs advance, the stable loss values suggest that the adversarial play between the generator and discriminator reaches a dynamic equilibrium. This implies the generator's proficiency in producing data that are indistinguishable from the real dataset and the discriminator's adeptness in classification.

To mitigate mode collapse and ensure stable convergence in our CGAN or QCGAN training, we adopted a Minibatch Discrimination technique along with the adaptive GAN approach [45]. This was implemented to simultaneously train ten distinct CGANs or QCGANs, effectively addressing the mode-collapse challenge [45]. It is important to note that the outputs referenced in the subsequent sections of this paper are derived through random selection, utilising a uniform distribution, from these ten CGANs or QCGANs.

TABLE 1 Test parameters.

Parameter	Value	Parameter	Value
Input dimension	3×1	Batch size	32
Output dimension	1	Learning rate	0.001
Condition qubits	3×8	Depth	2
Simulation qubits	8	Hidden dimension	2

4.1.6 | Hyperparameter and implementation details

The specific hyperparameters used in the various methods are listed in Table 1. For the neural network-based components of our model, we chose a batch size of 32 to balance the trade-off between computational efficiency and the ability to generalise. The learning rate is set to 0.001, which was selected carefully to ensure stable and efficient convergence during the training phase. A depth of two and a hidden dimension of two were employed to maintain model simplicity while capturing the necessary features, thus preventing overfitting.

The input dimension is 3×1, reflecting historical loss rate data spanning the past three days. This time frame was selected to capture recent market trends while avoiding the inclusion of outdated information that could potentially skew the model's predictions. Each daily loss rate is encoded using eight qubits, resulting in 3×8 qubits for the condition. This encoding depth was selected to provide a detailed representation of the loss rates while maintaining computational feasibility. Furthermore, the output dimension was set to one, represented by eight qubits, to predict the future daily loss rate. This choice is consistent with the aim of forecasting short-term market movements, thus allowing timely risk estimation. Therefore, the total number of coded segments was 2^8, ensuring a comprehensive yet manageable data representation within the quantum computing framework.

In response to the challenges posed by quantum noise, our circuit design was meticulously optimised. We prioritised minimising the circuit depth and reducing the number of quantum gates [46]. This approach significantly reduces the susceptibility of the circuit to noise, thereby improving the fidelity and reliability of computations. To rigorously assess the impact of the parameter choices on the performance of the framework, we conducted a comprehensive parameter sensitivity analysis, as detailed in Section IV.F.

4.2 | Accuracy comparison of optimal result

We compared the proposed QCGAN-QAE with existing QCGAN, CGAN, and historical simulation models to validate its effectiveness. Historical simulation methods are widely used as non-parametric market risk estimation methods. This comparison is based on their ability to accurately estimate the VaR and CVaR of the carbon market.

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TABLE 2 Comparison of accuracy between historical simulation, CGAN, QCGAN and QCGAN-QAE.

Model		Historical simulation			CGAN			QCGAN			QCGAN-QAE		
Confidence level		90%	95%	99%	90%	95%	99%	90%	95%	99%	90%	95%	99%
VaR	N (fail num.)	**28**	**18**	6	34	20	**3**	54	48	35	**28**	**18**	**3**
	r (%)	**9.06%**	**5.83%**	1.94%	11.00%	6.47%	**0.97%**	17.48%	15.53%	11.33%	**9.06%**	**5.83%**	**0.97%**
	Δr (%)	**0.94%**	**0.83%**	0.94%	1.00%	1.47%	**0.03%**	7.48%	10.53%	10.33%	**0.94%**	**0.83%**	**0.03%**
	LR	**0.3112**	**0.4217**	2.1708	0.3358	1.2958	**0.0027**	16.0628	47.4754	109.5332	**0.3112**	**0.4217**	**0.0027**
CVaR	Loss mean (fail.)	**5.4453**	6.3548	8.6980	5.1126	4.8289	**11.8597**	4.1407	4.3693	4.9903	5.4800	**6.4599**	9.2344
	CVaR mean (fail.)	**4.9024**	5.6113	6.2225	3.3847	7.4496	**9.9498**	2.4542	2.6459	3.0369	3.9583	**6.1182**	16.0058
	DLC	**0.5429**	0.7435	2.4754	1.7279	2.6208	**1.9099**	1.6866	1.7234	1.9534	1.5217	**0.3417**	6.7714

Abbreviations: CGAN, conditional generative adversarial network; CVaR, conditional value at risk; DLC, difference between the losses and CVaR; LR, likelihood ratio; QCGAN, quantum conditional generative adversarial network; r, failure ratio; VaR, value at risk.

For the historical simulation, we selected a rolling time window comprising the most recent 100 trading days to calculate the daily VaR and CVaR, in line with commonly accepted risk management practices [47]. Each day, the window moves forward to include the latest available return, while excluding the oldest, ensuring that the model remains responsive to the most current market information. This rolling window approach captures temporal variations in the data and provides dynamic and up-to-date risk estimation.

Direct measurement of the QCGAN output is a crucial baseline model. Once the QCGAN generates a quantum state that encodes the probability distribution of the potential future loss rates, we measure this state to derive concrete risk estimates. The measurement involves collapsing the quantum state into classical information, which essentially translates the probabilities encoded in the quantum state into observable outcomes.

Table 2 compares the accuracy of the historical simulation, CGAN, QCGAN, and QCGAN-QAE. We found that the LR value of the original QCGAN is too large and exceeded the critical value; therefore, it failed to pass the Kupiec test at all three confidence levels. Therefore, the estimation ability of the QCGAN could not reflect the real VaR and CVaR situations, and we only analysed the historical simulation, CGAN, and QCGAN-QAE that passed the Kupiec test as follows: We found that, at a 90% confidence level, the historical simulation method and QCGAN-QAE are significantly better than QCGAN. Among them, the historical simulation method and QCGAN-QAE had the same performance on VaR, whereas their performance on CVaR is slightly better than that of the historical simulation method. The situation at the 95% confidence level was similar; both the historical simulation and QCGAN-QAE are better than the CGAN. In terms of VaR performance, the historical simulation and QCGAN-QAE performances are still the same; however, it should be noted that QCGAN-QAE at the 95% confidence level performs better in CVaR. At the 99% confidence level, the historical simulation method performs poorly, whereas the CGAN and QCGAN-QAE perform better. Among them, the CGAN and QCGAN-QAE performed similarly on VaR, whereas the CGAN performed better on CVaR. Overall, a comparison between the historical simulation, CGAN, QCGAN, and QCGAN-QAE shows that the proposed model generally has the best overall effect, with outstanding performance at the three confidence levels. While the historical simulation method performed poorly at the 99% confidence level, the CGAN performed poorly at the 90% and 95%, especially at 90% confidence levels. This finding shows that, although QCGAN-QAE performs well under normal market conditions, there are certain limitations in its ability to face extreme market risks.

The poor prediction accuracy of the QCGAN may be attributed to several key factors. First, the existing QCGAN requires a more complex quantum circuit design, and the sequence of conditional data interactions and data simulations hinders effective quantum information exchange. Second, the method relies only on direct measurements, and errors can easily occur when measuring low-probability extreme values. The data in Table 2 also verify this phenomenon; while QCGAN shows a relatively strong performance at the 90% confidence level, its prediction performance drops sharply at the 99% confidence level.

The reason QCGAN-QAE performs worse than the classical CGAN at the 99% confidence level may be the limitation of qubits. In our proposed QCGAN-QAE model, the number of qubits used is directly related to the accuracy of the prediction; therefore, there may still be some limitations in the accuracy of extreme values, resulting in a 99% confidence level CVaR estimation error. As quantum computers develop and an increasing number of qubits become available, this limitation progressively diminishes.

Generally, this comparison shows that QCGAN-QAE has improved accuracy in most cases compared to the traditional historical simulations, CGAN, and QCGAN estimation results, illustrating the huge potential of quantum algorithms in carbon market risk estimation.

4.3 | Computational efficiency comparison of optimal result

To compare the advantages of the quantum algorithm QCGAN-QAE in terms of computational efficiency, we compared the running time and epochs required for the convergence

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TABLE 3 Comparison of computational efficiency between CGAN and QCGAN-QAE.

	CGAN			QCGAN-QAE		
	90%	95%	99%	90%	95%	99%
Quantum computer (s)	/	/	/	**0.0020**	**0.0020**	**0.0021**
Traditional computer (s)	169	168	167	**992**	**990**	**994**
Epoch	300	300	300	**30**	**30**	**30**

Abbreviations: CGAN, conditional generative adversarial network; QCGAN-QAE, quantum conditional generative adversarial network-quantum amplitude estimation.

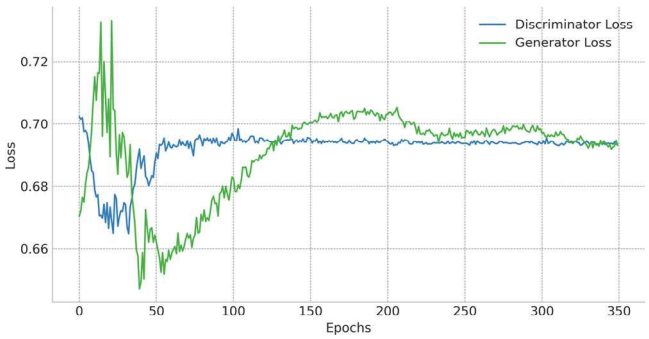

FIGURE 5 Discriminator and generator losses in CGAN training.

of the proposed QCGAN-QAE with the traditional CGAN, which is similar to the accuracy.

Table 3 presents the results of the various time indicators required to execute each model of carbon market risk estimation. We compare the running time of quantum algorithms on quantum computing with that of traditional algorithms on traditional computers. In particular, when the confidence levels are 90% and 95%, QCGAN-QAE is slightly more accurate than CGAN, and the required running time is only 1/84,500 (a reduction of 99.9988%) and 1/84,000 of CGAN (a reduction of 99.9988%). Even when the confidence level is 99%, QCGAN-QAE has a significant advantage when the accuracy is slightly worse than that of the CGAN, and the required running time is only 1/79,524 of that of the CGAN (a reduction of 99.9987%). Therefore, comparing the running time of classical algorithms on classical computers and the running time of quantum algorithms on quantum computers, quantum algorithms have strong advantages. For the epochs required for convergence, QCGAN-QAE requires only 30 epochs to achieve convergence, and the required epochs are reduced by 90% compared to the 300 epochs required by the CGAN, which means that the quantum algorithm has the advantage of convergence. Overall, our analysis indicates that the quantum parallelism inherent in QCGAN-QAE significantly enhances the computational speed. However, the increased time at the 99% confidence level suggests the need for further optimisation of quantum circuitry to handle complex risk scenarios more efficiently.

Figures 5 and 6 present a comparison between the convergence behaviours of CGAN and QCGAN-QAE. This comparison highlights the superior convergence efficiency of QCGAN-QAE, which is characterised by its accelerated stabilisation and reduced oscillation in loss values. The advantage of the quantum model convergence performance may arise from quantum parallelism and optimisation capabilities. These findings highlight the potential of the QCGAN-QAE for achieving faster and more stable convergence, signifying the great potential of quantum algorithms for complex tasks.

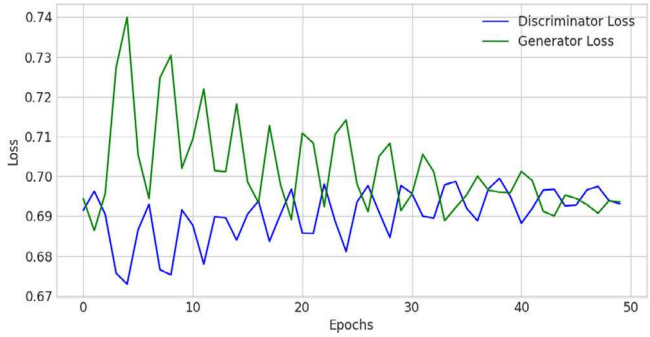

FIGURE 6 Discriminator and generator losses in QCGAN-QAE training.

The computational efficiency of QCGAN-QAE exceeds that of CGAN, which may be attributed to quantum parallelism and a more favourable optimisation environment. Quantum parallelism enables a QCGAN to perform multiple computations simultaneously, speeding up efficiency, whereas a CGAN on classical computers can only perform sequential operations. Furthermore, quantum computers can more effectively navigate complex, potentially non-convex GAN optimisation environments, enabling the QCGAN to find more favourable convergence paths. Therefore, the outstanding computational speed of the QCGAN-QAE is owing to its ability to perform multiple computations simultaneously and traverse complex optimisation environments more efficiently.

In general, this comparison demonstrates the advantages of the proposed QCGAN-QAE algorithm in terms of computational efficiency when the accuracy is approximately equivalent to or better than that of a traditional algorithm.

4.4 | Accuracy comparison of QCGANs with different quantum circuit structures

This experiment aims to evaluate the performance of the proposed QCGAN algorithm, particularly to compare different quantum circuit structures and their accuracy in estimating the carbon market risk. It is crucial to understand how different circuit architectures affect the results and to choose the most suitable structure for practical applications. The considered structures included the original, proposed, and proposed quantum circuits with additional QFC.

Table 4 presents an accuracy comparison of QCGANs with different quantum-circuit structures. The results show that the estimation accuracy of the modified QCGAN is significantly higher than that of the original QCGAN. Since the LR value of the original QCGAN is excessively large and exceeds the critical value, it cannot pass the Kupiec test at the 90%, 95%, and 99%

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TABLE 4 Comparison of accuracy between QCGANs with different quantum circuit structures.

Model		QCGAN (original)			QCGAN (modified)			QCGAN (modified+QFC)		
Confidence level		90%	95%	99%	90%	95%	99%	90%	95%	99%
VaR	N (fail num.)	54	48	35	44	19	**4**	**42**	**17**	4
	r (%)	17.48%	15.53%	11.33%	14.24%	6.15%	**1.29%**	**13.59%**	**5.50%**	1.29%
	Δr (%)	7.48%	10.53%	10.33%	4.24%	1.15%	**0.29%**	**3.59%**	**0.50%**	0.29%
	LR	16.0628	47.4754	109.5332	5.5291	0.8026	**0.2477**	**4.0298**	**0.1587**	0.2477
CVaR	Loss mean (fail.)	4.1407	4.3693	4.9903	4.5261	6.6463	**10.6796**	**4.6087**	**6.7394**	10.3159
	CVaR mean (fail.)	2.4542	2.6459	3.0369	3.1831	7.4391	**27.1200**	**3.6205**	**7.1417**	35.2923
	DLC	1.6866	1.7234	1.9534	1.3430	0.7927	**16.4404**	**0.9882**	**0.4023**	24.9763

Abbreviations: CVaR, conditional value at risk; DLC, difference between the losses and CVaR; LR, likelihood ratio; QCGAN, quantum conditional generative adversarial network; QFC, quantum fully connected layer; r, failure ratio; VaR, value at risk.

confidence levels; however, the modified model can pass the 95% and 99% confidence level tests. This suggests that quantum circuits that perform entanglement followed by rotation can better capture the distribution of the loss rate. Additionally, we compare the estimation accuracy of the model with and without QFC. At 90% and 95% confidence levels, those with QFC performed better on both VaR and CVaR. Specifically, at 90% confidence level, in terms of Δr, QCGAN-QAE decreased by 15.33%; in terms of DLC, QCGAN-QAE decreased by 26.42%. At 95% confidence level, in terms of Δr, QCGAN-QAE decreased by 56.52%; in terms of DLC, QCGAN-QAE decreased by 49.25%. At the 99% confidence level, although the effects of both VaRs were the same (QCGAN without QFC), the DLC is relatively small. However, it should be noted that although the DLC without QFC (16.4404) is still too large, the comparison between the two is not meaningful, and we will solve this problem in Section 4.5. In summary, the structural choices in the quantum circuit, which are the order of the entanglement and rotation stages and the inclusion of the QFC layer, significantly enhance the QCGAN's ability to accurately estimate market risks. However, the QCGAN with QFC is still slightly better.

The experimental results validate the rationale behind swapping the entanglement and rotation stages. The enhancement in accuracy brought about by the swap may stem from the strong quantum correlations established through entanglement prior to the rotation stage, which provides a more favourable condition for rotation operations. Consequently, this improves the expressiveness and efficiency of the quantum circuit, further increasing estimation accuracy. In addition, the advantage of adding a QFC is its ability to interact with information more effectively so that the information can be fully characterised, ultimately improving the estimation accuracy.

In conclusion, the experiments show that modifying the quantum circuit structure and adding a QFC can significantly enhance the performance of the QCGAN-QAE algorithm in terms of carbon market risk estimation. This improvement is critical for developing more accurate and efficient carbon market risk-management tools.

4.5 | Comparison of QAE and measure

This experiment aimed to demonstrate the advantages of QAE over classical measurement methods and highlight the improved time efficiency of combining QAE with binary search methods. In this section, we compare the performance of the QAE with that of direct classical measurement methods for estimating the amplitudes of quantum states generated by QCGAN models. Furthermore, we evaluated the computational efficiency obtained by integrating the binary search method into the QAE instead of using the QAE alone.

Table 5 presents the estimated performance of QCGAN with and without QAE. The results show that after adding the QAE, the estimation accuracy improves. When the confidence level is 90%, the LR of the QCGAN without the QAE was too large to pass the Kupiec test, whereas the QCGAN with the QAE passed the test. When the confidence level is 95%, we found that although the performance of QCGAN with QAE in the Δr has declined, it still passed the test successfully, and the performance of QCGAN-QAE in DLC has improved by 15.06%. The most important aspect is performance improvement at the 99% confidence level. In terms of Δr, QCGAN-QAE has been reduced by 89.66%, and in terms of DLC, it has been reduced by 72.89%. This solves the previously mentioned problem in which DLC have always been too large. Integrating a binary search with the QAE significantly improved the time efficiency of the model. This combination is particularly effective in accurately estimating the probabilities of extreme loss rates, which is a crucial aspect of risk management in volatile markets such as the carbon market.

The reason QAE can improve prediction accuracy and solve the CVaR prediction problem with 99% confidence may be the nature of the QAE algorithm itself. The QAE is a vital quantum algorithm for estimating the amplitude of quantum states, which is crucial for various quantum computing applications. The QAE improves the accuracy of predictions by amplifying the amplitude of a given quantum state. This property has critical implications for measuring quantum states with small probabilities, that is, the probability of extreme loss rates that

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TABLE 5 Compare the results of QCGAN combined with QAE on estimation accuracy.

Model		QCGAN (original)			QCGAN-QAE		
Confidence level		90%	95%	99%	90%	95%	99%
VaR	N (fail num.)	42	**17**	4	**28**	18	**3**
	r (%)	13.59%	**5.50%**	1.29%	**9.06%**	5.83%	**0.97%**
	Δr (%)	3.59%	**0.50%**	0.29%	**0.94%**	0.83%	**0.03%**
	LR	4.0298	**0.1587**	0.2477	**0.3112**	0.4217	**0.0027**
CVaR	Loss mean (fail.)	4.6087	6.7394	10.3159	**5.4800**	**6.4599**	**9.2344**
	CVaR mean (fail.)	3.6205	7.1417	35.2923	**3.9583**	**6.1182**	**16.0058**
	DLC	0.9882	0.4023	24.9763	**1.5217**	**0.3417**	**6.7714**

Abbreviations: CVaR, conditional value at risk; DLC, difference between the losses and CVaR; LR, likelihood ratio; QCGAN, quantum conditional generative adversarial network; QCGAN-QAE, quantum conditional generative adversarial network-quantum amplitude estimation; r, failure ratio; VaR, value at risk.

TABLE 6 Comparison of computational time combined with binary search.

Model	Confidence level	Quantum computer (s)
QCGAN-QAE	90%	0.0122
	95%	0.0064
	99%	0.0017
QCGAN-QAE (binary search)	90%	**0.0020**
	95%	**0.0020**
	99%	**0.0021**

Abbreviation: QCGAN-QAE, quantum conditional generative adversarial network-quantum amplitude estimation.

occur in our case. By contrast, direct measurement is a simple technique that often requires multiple repetitions to obtain precise results.

Table 6 presents a comparison of the computational time combined with a binary search. By comparing the results, we found that at the 90% confidence level, the QAE efficiency combined with the bisection method reduced the running time on the quantum computer by 83.61%, and at the 95% confidence level, the running time was shortened by 68.75%. The running time increased by 23.53% at a 99% confidence level. In general, when the confidence level is low, acceleration combined with the binary search method is more obvious. When the confidence level is particularly high, the QAE may only be as good as the binary search method.

The efficiency improvement of the QAE originates from the binary search. Binary search is a classic algorithm. Combining the binary search method with the QAE accelerates the computational process by reducing the number of operations required. The time complexity of the QAE combined with the binary search method is reduced from O(N) to O(logN). As for the 99% confidence level, it takes longer to combine the binary search because the original QAE is measured from the quantum state that represents the largest loss rate, and VaR with a 99% confidence level is generally a large loss rate; therefore, fewer measurements may be required.

In conclusion, the experiments demonstrated that integrating the binary search and QAE can dramatically accelerate the computation process. This improvement has important implications for quantum-computing applications, particularly risk estimation.

4.6 | Parameter sensitivity analysis

Generally, the performance of a robust QCGAN-QAE is relatively high when the depth fluctuates within a reasonable range. To demonstrate the universality of the QCGAN-QAE predictions, the QCGAN-QAE models were compared at depths of 2, 4, and 6.

Table 7 indicates that, within the selected depth range, the performance of the model is not sensitive to the depth choice. This insensitivity may be attributed to our modified QCGAN and the replacement of the simple measure with QAE. In particular, the refined QCGAN ensures that the essential properties of the quantum circuit do not change significantly with depth. Moreover, using the QAE helps obtain more accurate estimates irrespective of the circuit depth. Our sensitivity analysis suggests that the QCGAN-QAE model maintains robust performance across a range of circuit depths. This resilience is vital for ensuring the reliability and accuracy of the model, regardless of the available quantum computing resources.

Additionally, it is important to note that although increasing the depth of a quantum circuit can lead to more expressive models, it also increases the risk of overfitting and the impact of noise. The chosen range of depths seems optimal for balancing the tradeoff between model efficiency and accuracy.

5 | CONCLUSION

Carbon market risk estimation plays an important role in global environmental governance and strategic corporate decision making. To satisfy the carbon market's demand for higher accuracy and efficiency in risk estimation, we introduced the QCGAN-QAE framework. By combining the estimation

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TABLE 7 Results of VaR/CVaR estimated by QCGAN-QAE with different depths.

Model		2			4			6		
Confidence level		90%	95%	99%	90%	95%	99%	90%	95%	99%
VaR	N (fail num.)	**28**	**18**	**3**	25	18	2	34	19	2
	r (%)	**9.06%**	**5.83%**	**0.97%**	8.09%	5.83%	0.65%	11.00%	6.15%	0.65%
	Δr (%)	**0.94%**	**0.83%**	**0.03%**	1.91%	0.83%	0.35%	1.00%	1.15%	0.35%
	LR	**0.3112**	**0.4217**	**0.0027**	1.3303	0.4217	0.4438	0.3358	0.8026	0.4438
CVaR	Loss mean (fail.)	5.4800	6.4599	9.2344	5.7935	6.4597	13.9113	**4.6143**	**5.2242**	**13.9113**
	CVaR mean (fail.)	3.9583	6.1182	16.0058	4.3859	5.3343	25.2832	**3.6005**	**5.4312**	**9.9307**
	DLC	1.5217	0.3417	6.7714	1.4076	1.1254	11.3720	**1.0137**	**0.2070**	**3.9806**

Abbreviations: CVaR, conditional value at risk; DLC, difference between the losses and CVaR; LR, likelihood ratio; r, failure ratio; VaR, value at risk.

accuracy of QCGAN with the computational efficiency of QAE, this novel quantum computing-based framework significantly advances carbon market risk estimation. Specifically, the improvements we made to the circuit structure of the QCGAN led to a notable increase in the accuracy of VaR and CVaR estimation. Furthermore, integrating the binary search method with QAE not only improves the risk estimation accuracy with limited measurements but also considerably enhances the model efficiency.

From an application perspective, the simulation results of our QCGAN-QAE framework provide insights into enhancing carbon market risk management strategies. The increased accuracy of risk estimation can inform the development of more effective regulatory policies and market-oversight mechanisms. This could lead to the adoption of dynamic risk-management approaches that respond to real-time market data, aiming to stabilise the carbon market and reduce systemic risks.

Empirical evidence demonstrates that the QCGAN-QAE framework improves the estimation accuracy compared to the historical simulation, classical CGAN, and original QCGAN. Although the primary application of our QCGAN-QAE framework is in the area of carbon market risk, its potential extends to a broader range of risk estimation applications across various sectors. Our research focuses on minimising the dependency on the qubit count. To achieve this, we plan on exploring the use of angular rotation encoding, which promises to significantly refine the approach to quantum data encoding. In addition, there will an emphasis on fine-tuning the algorithm parameters of the QCGAN-QAE framework. The aim is to develop more effective strategies for parameter selection and adjustment to enhance the overall performance of the model. This may include the use of advanced optimisation techniques and the potential incorporation of machine-learning algorithms to dynamically adjust parameters in response to changing market conditions.

NOMENCLATURE

Abbreviations

QCGAN	Quantum conditional generative adversarial network
CGAN	Conditional generative adversarial network
QAE	Quantum amplitude estimation
QCGAN-QAE	Quantum conditional generative adversarial network-quantum amplitude estimation
VaR	Value at risk
CVaR	Conditional value at risk
PQC	Parameterized quantum circuit
QFC	Quantum fully connected layer
CNOT	Controlled NOT gate
DLC	Differences between losses and CVaR
CDF	Cumulative distribution function
EU ETS	European Union Emissions Trading System
ECX	European Climate Exchange

Parameters

n	Number of qubits
$D_{\mathrm{QCGAN},i}$	Circuit depth for QCGAN
$D_{\mathrm{QAE},i}$	Circuit depth for QAE
T	Total number of observations
T_P	Time to prepare the initial state
T_G	Average duration of a quantum gate
T_M	Time needed to measure the qubits
α	Given confidence level
p	Theoretical failure ratio, that is 1-α

Variables

x	Market object quantum data
y	Market condition quantum data
z	Quantum noise vector
α	Amplitude
θ	Angle
p_{QAE}	Amplitude measured using quantum amplitude estimation
y_t	Historical loss rate at time t
R_t	Rate of return at time t
p_t	Carbon market price at time t
T_i	Unified computing time
T_{total}	Cumulative computing time
T_P	Time to prepare the initial state
N	Number of failures (times actual loss rate exceeds VaR)

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r	Failure ratio
Δr	Discrepancy measure
X_i	The real loss rate on day i
CVaR_i	CVaR on day i
L	Loss rate obtained after decoding the quantum state set
$f(L)$	Corresponding probability of loss rate L
VaR_α	VaR under confidence level α
CVaR_α	CVaR under confidence level α

ACKNOWLEDGEMENTS

This work was supported by the National Natural Science Foundation of China, 72331009, 72171206; Guangdong Power Grid Company, 036000KC23090003(GDKJXM20231024); Shenzhen Key Lab of Crowd Intelligence Empowered Low-Carbon Energy Network, ZDSYS20220606100601002; Shenzhen Natural Science Fund (the Stable Support Plan Program GXWD20231128112434001); Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS).

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data supporting this study's findings are available in Intercontinental Exchange (ICE). The website is https://www.ice.com/index.

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How to cite this article: Zhou, X., Zhao, H., Cao, Y., Fei, X., Liang, G., Zhao, J.: Carbon market risk estimation using quantum conditional generative adversarial network and amplitude estimation. Energy Convers. Econ. 5, 193–210 (2024). https://doi.org/10.1049/enc2.12122

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